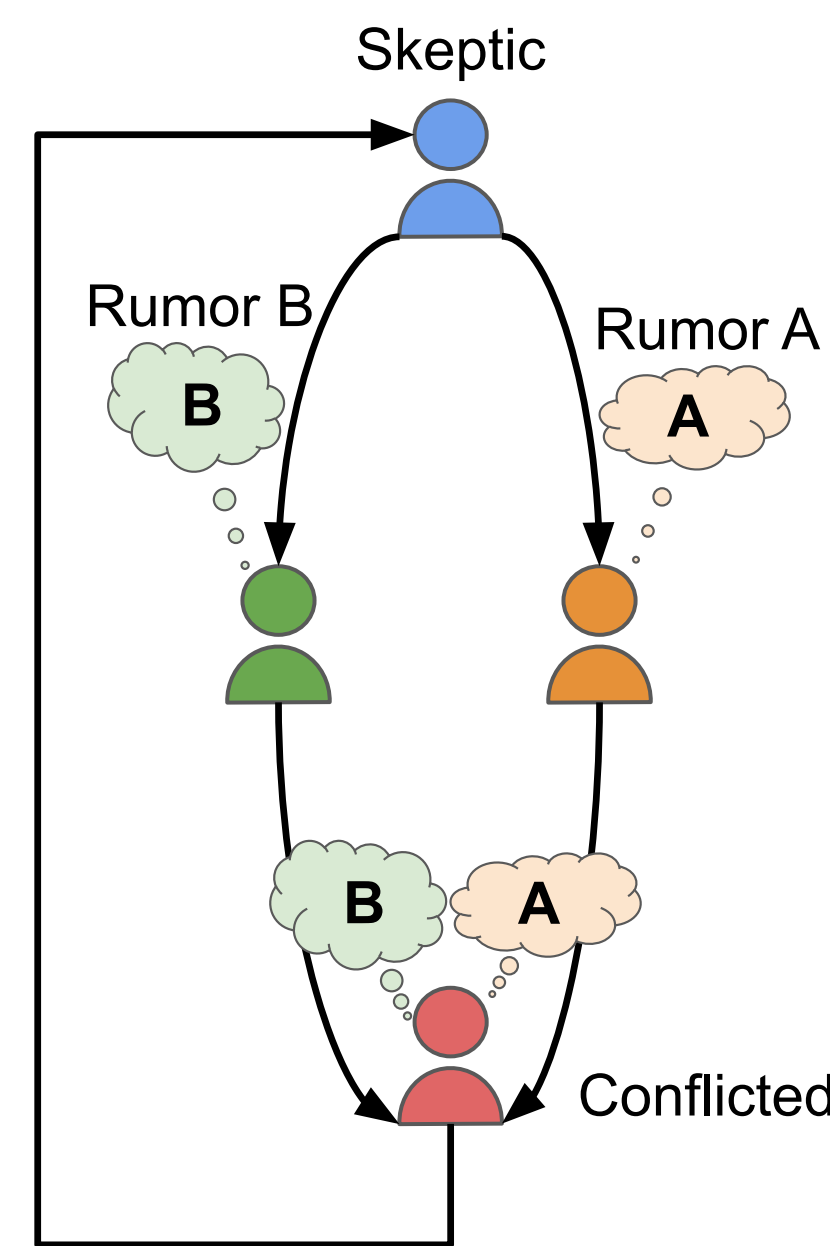


Social media platforms bombard users with vast amounts of information, often exceeding individual cognitive processing capacities. In contexts where forming an informed opinion is costly, individuals tend to rely on social reinforcement, validating information only if it is confirmed by their local environment. In this work, we study the propagation of unverified information (rumors) in complex networks, focusing on a scenario with two competing rumors regarding a single topic: one positive and one negative. We introduce a mechanism where these rumors are correlated, such that the adoption of one facilitates the adoption of the other. However, since the rumors represent opposing views, agents who accept both encounter a state of cognitive dissonance. This conflict is eventually resolved by discarding both beliefs and adopting a stricter criterion (higher threshold) for accepting future information. Our model revealed two distinct parameter regions: one where a single rumor dominates, and another where most agents become skeptical. A proposed dynamical description, when integrated, showed similar behavior to the original model.

Model description

- Agent-based model constructed on a **complex network**.
- Agents have an **adoption threshold** φ_i , that represents the fraction of its neighbors that are required to believe in the rumor for them to also adopt it.
- There are two rumors which agents could adopt, or “be activated in”, one positive and one negative about a subject.
- Rumors can be correlated, meaning that agents that believe one rumor can influence in the activation of the other. The parameter λ indicates the degree of correlation between both rumors.
- Believing in both rumors generates a certain indecision in the agent. To fix this, agents deactivate in both rumors with probability α , which represents the intensity of the dissonance.
- When an agent discards both rumors, its adoption threshold φ_i increases by a fixed amount δ .

Agent's states



Rules for rumor adoption

$$f_A^{ef} = \frac{\text{orange} \times 1 + \text{green} \times \lambda + \text{red} \times (1 + \lambda)}{\text{orange} + \text{green} + \text{red}} = 0.6$$

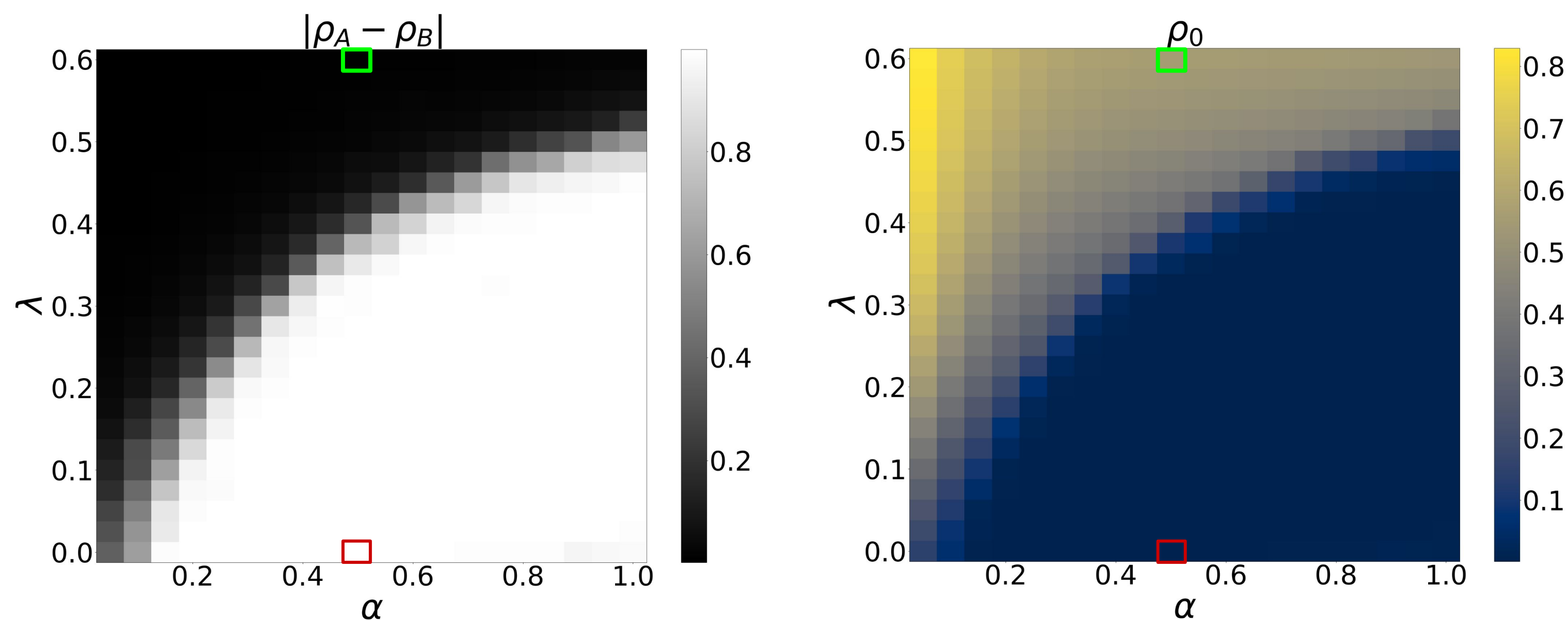
$$f_B^{ef} = \frac{\text{green} \times 1 + \text{orange} \times \lambda + \text{red} \times (1 + \lambda)}{\text{orange} + \text{green} + \text{red}} = 0.4$$

$$f_B^{ef} = \frac{\text{green} \times 1 + \text{orange} \times \lambda + \text{red} \times (1 + \lambda)}{\text{orange} + \text{green} + \text{red}} = 1$$

Simulation results

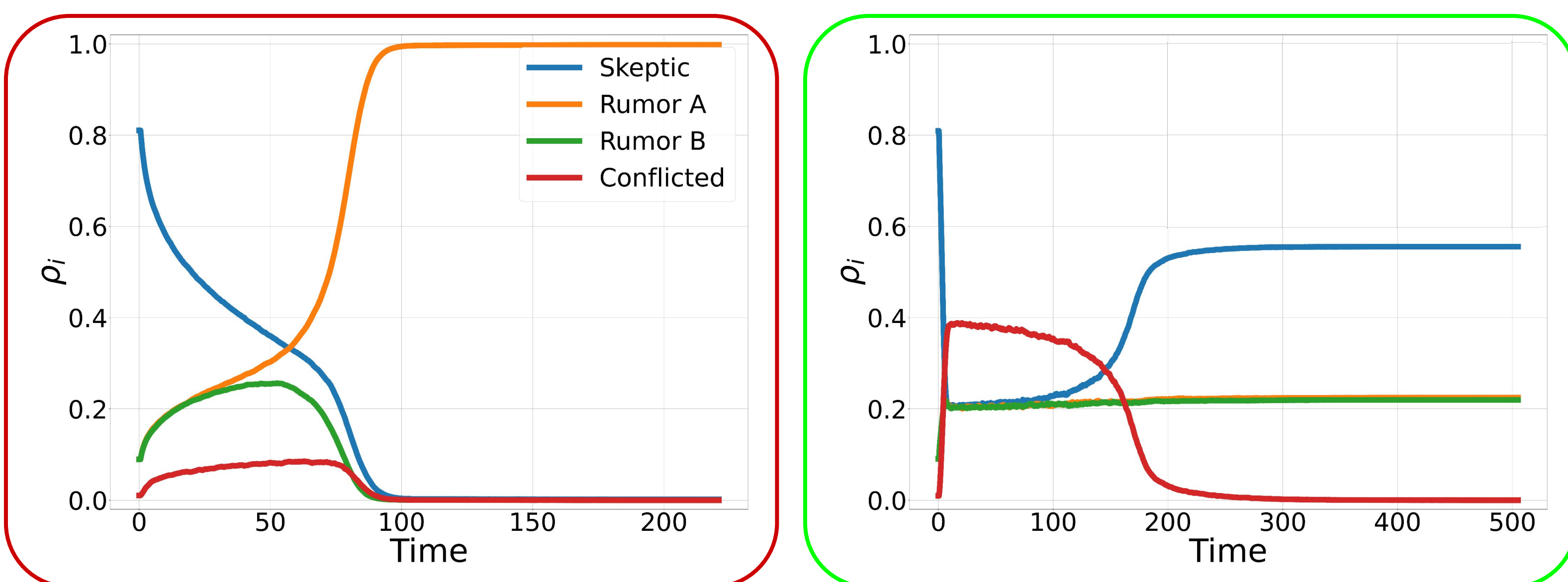
Parameters space:

Agents cannot remain in the Conflicted state, so at the end of the simulation they either believe in one rumor or they are skeptic. To characterize the effect of the parameters λ and α we simulated the model for a fixed value $\delta=0.01$ using an Erdős-Rényi Network of size $N=100.000$, $\langle k \rangle=6$ and with a 100 simulations per point in the parameter space.



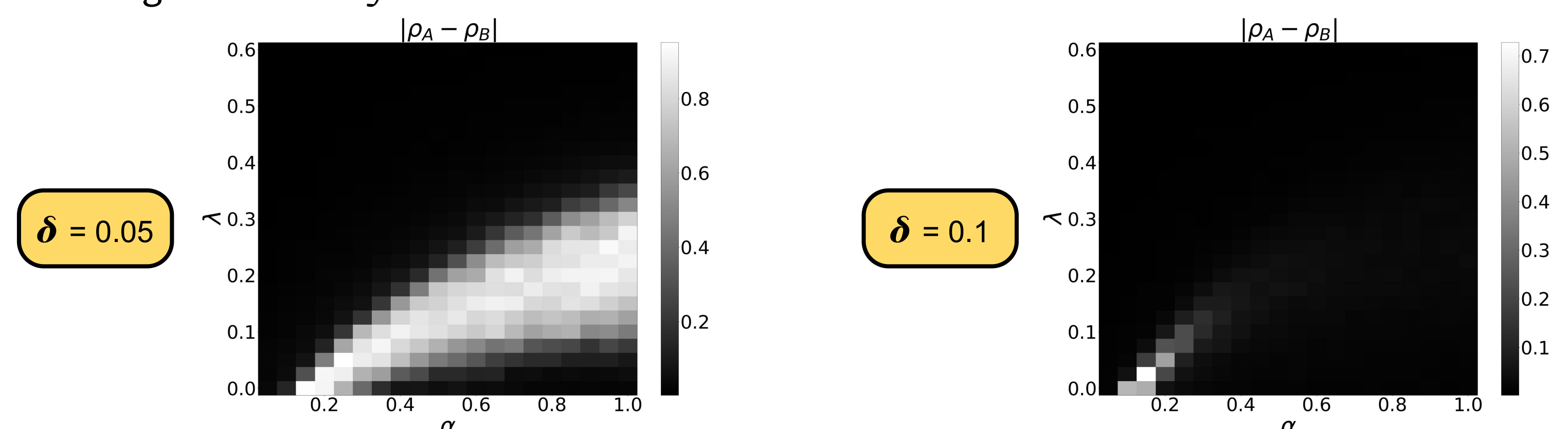
Final states:

Two regions emerge in the parameter space. We show an example of the dynamics of each region, one in which one rumor dominates the network, the other in which the majority of agents end up being Skeptic. We plot the fraction of agents in each state as a function of time.



Stricter agents:

The parameter δ determines how much stricter an agent becomes after discarding both rumors. The higher it is, the less deactivations an agent requires to be unable to believe in any rumor again. We study its effect on the behavior of the model.



Homogeneous mean field approximation

Dynamic equations:

We consider that each agent has the same vicinity and therefore, that the fraction of neighbors in one state is the same as the fraction of agents in the network in that state.

$$\frac{d\rho_0}{dt} = -\omega_A \rho_0 [\rho_A + \lambda \rho_B + (1 + \lambda) \rho_{A,B}] - \omega_B \rho_0 [\rho_B + \lambda \rho_A + (1 + \lambda) \rho_{A,B}] + \alpha \rho_{A,B}$$

$$\frac{d\rho_A}{dt} = \omega_A \rho_0 [\rho_A + \lambda \rho_B + (1 + \lambda) \rho_{A,B}] - \omega_B \rho_A [\rho_B + \lambda \rho_A + (1 + \lambda) \rho_{A,B}]$$

$$\frac{d\rho_B}{dt} = \omega_B \rho_0 [\rho_B + \lambda \rho_A + (1 + \lambda) \rho_{A,B}] - \omega_A \rho_B [\rho_A + \lambda \rho_B + (1 + \lambda) \rho_{A,B}]$$

$$\frac{d\rho_{A,B}}{dt} = \omega_B \rho_A [\rho_B + \lambda \rho_A + (1 + \lambda) \rho_{A,B}] + \omega_A \rho_B [\rho_A + \lambda \rho_B + (1 + \lambda) \rho_{A,B}] - \alpha \rho_{A,B}$$

$$\omega_A(t) = \int_{-\infty}^{f_A^{ef}} \Phi(\varphi, t) d\varphi \quad f_A^{ef} = \rho_A + \lambda \rho_B + (1 + \lambda) \rho_{A,B}$$

$$\omega_B(t) = \int_{-\infty}^{f_B^{ef}} \Phi(\varphi, t) d\varphi \quad f_B^{ef} = \rho_B + \lambda \rho_A + (1 + \lambda) \rho_{A,B}$$

ρ_s is the fraction of agents in the state s .

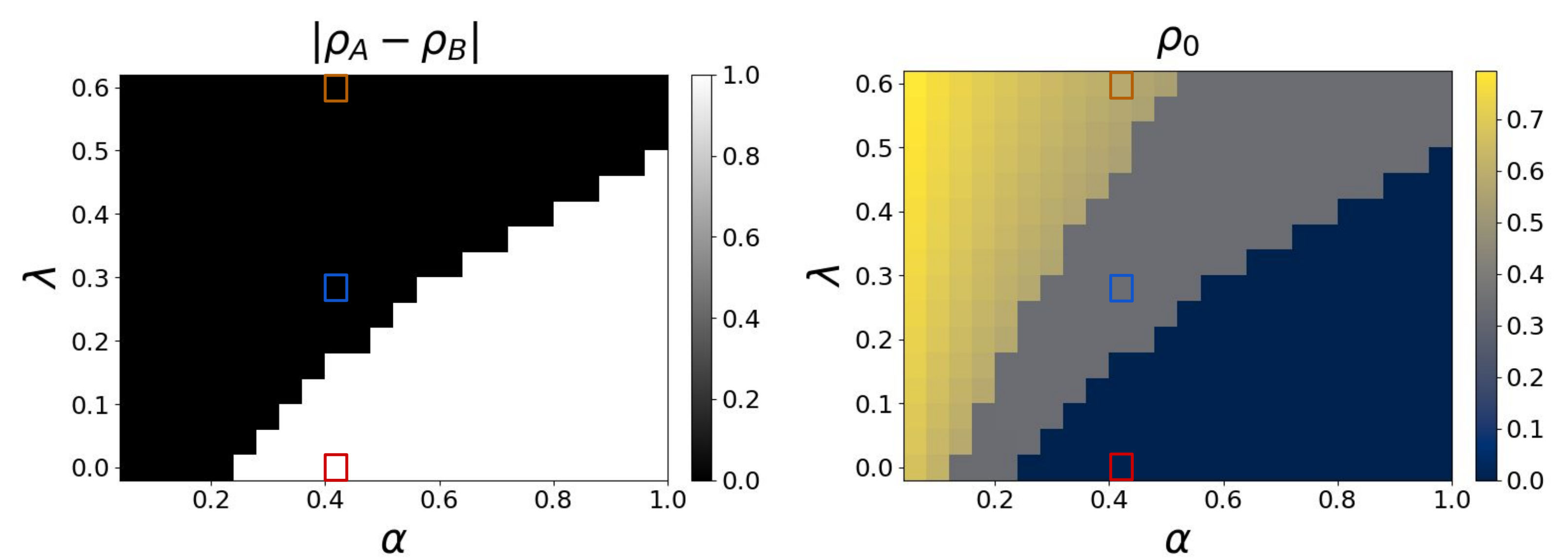
ω_A is the fraction of agents with thresholds lower than the effective fraction of neighbors that believe in rumor A. ω_B is similar but for rumor B.

f_A^{ef} is the effective fraction of neighbors that believe in rumor A. f_B^{ef} is the same for rumor B.

$\Phi(\varphi, t)$ is the time dependent distribution of thresholds.

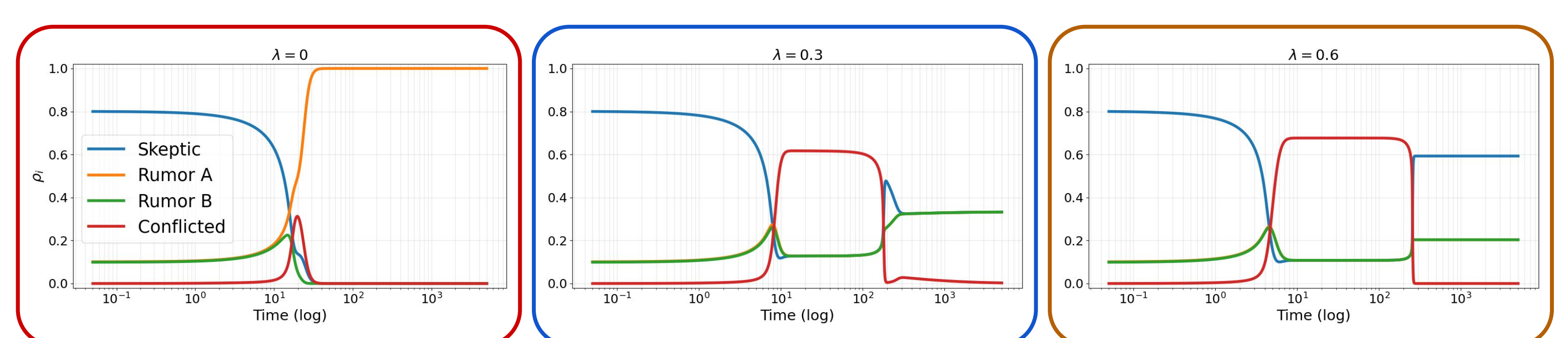
Integrated simulations:

We integrate our dynamical equations using a RK4 method to compare our theoretical description with the model. We simulate this for a fixed value $\delta=0.01$.



Final states:

We plot the fraction of agents in each state as a function of time in the three regions observed.



References

- “A simple model of global cascades on random networks”, D.J. Watts. Proc. Natl. Acad. Sci. U.S.A. 99 (9) 5766-5771, DOI: 10.1073/pnas.082090499 (2002)
- “Thresholds models of collective behavior”, M. Granovetter. American Journal of Sociology, 83, DOI: 10.1086/226707 (1978)